

# 2016 HKDSE Physics

## Paper 1B Structured Questions

Original questions with official answers and English explanations. Correct answers are red; explanations are blue.

An asterisk marks extension content in the original paper.

### Question 1

The following experimental items are provided for estimating the specific heat capacity of bronze  $c_b$ :

a bronze sphere of mass 0.80 kg hung with a thread at room temperature  $T_0$

a polystyrene cup containing 0.50 kg of water at room temperature  $T_0$

a water bath maintained at 80 °C

a thermometer

a stirrer

a towel

(a) Describe the procedures of the experiment and state TWO experimental precautions to be taken. Write down an equation for finding  $c_b$ .

Given: specific heat capacity of water = 4200 J kg<sup>-1</sup> °C<sup>-1</sup>. (6 marks)

(b) The value of  $c_b$  found in the experiment in (a) is lower than the actual value. Explain. (2 marks)

### Official marking scheme

| Solution                                                                                                                            | Marks | Remarks |
|-------------------------------------------------------------------------------------------------------------------------------------|-------|---------|
| 1. (a) - Put the sphere into the water bath for a few minutes                                                                       | 1A    |         |
| - Put / transfer the sphere into the polystyrene cup (of water)                                                                     | 1A    |         |
| - Measure the final / maximum temperature $T_f$ of the water with a thermometer                                                     | 1A    |         |
| $0.80 \times c_b \times (80 - T_f) = 0.50 \times 4200 \times (T_f - T_0)$                                                           | 1A    |         |
| $c_b = 2625 \times \frac{T_f - T_0}{80 - T_f}$ (J kg <sup>-1</sup> °C <sup>-1</sup> )                                               |       |         |
| Precautions:                                                                                                                        |       |         |
| - Dry the sphere with the towel quickly before putting it into the cup                                                              | 1A    |         |
| - Make sure the sphere is fully immersed in water                                                                                   | 1A    |         |
| - Stir the water thoroughly                                                                                                         | 1A    |         |
|                                                                                                                                     | 6     |         |
| (b) Thermal energy / heat is lost during the transfer / drying of the sphere                                                        | 1A    |         |
| Or Thermal energy / heat absorbed by thermometer, stirrer or cup                                                                    | 1A    |         |
| Or The temperature of the sphere is higher than $T_f$ when this final temperature is measured (i.e. $T_f$ not reaching its maximum) | 1A    |         |
| Thus temperature rise of water in the cup is lower than it should be.                                                               | 1A    |         |
|                                                                                                                                     | 2     |         |

(a) Measure the initial water temperature  $T_0$ . Leave the sphere in the  $80\text{ }^{\circ}\text{C}$  bath for several minutes so it reaches the bath temperature. Transfer it into the water in the polystyrene cup. Stir and record the maximum/final common temperature  $T_f$ .

Assuming negligible heat loss, heat lost by bronze equals heat gained by water:

$$0.80c_b(80 - T_f) = 0.50 \times 4200(T_f - T_0).$$

$$\text{Therefore } c_b = 2625(T_f - T_0)/(80 - T_f) \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}.$$

Precaution 1: quickly dry the sphere before transfer so hot bath water does not enter the cup.

Precaution 2: fully immerse the sphere and stir the water thoroughly so the recorded temperature represents the mixture.

(b) Heat is lost while the sphere is dried/transferred, or is absorbed by the cup, thermometer and stirrer. The measured water-temperature rise is therefore smaller than assumed in the ideal heat balance, giving a value of  $c_b$  that is too low.

## Question 2 \*

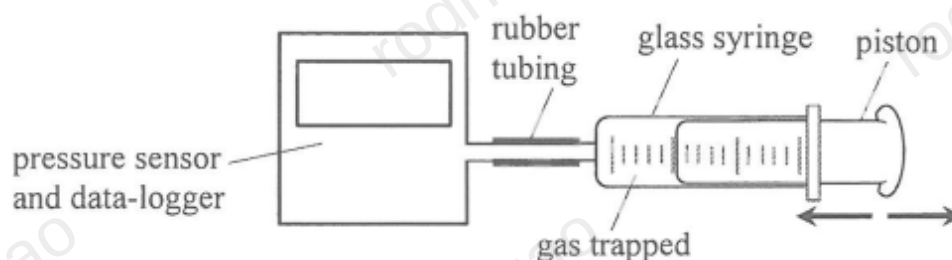


Figure 2.1

Judy uses the set-up shown in Figure 2.1 to study the relationship between the pressure and volume of a fixed mass of gas at constant temperature. The volume  $V$  of the gas trapped is read directly from the syringe and the corresponding pressure  $p$  is measured by a data logger via a pressure sensor.

(a) The initial volume and pressure of the gas are  $6.0 \times 10^{-5} \text{ m}^3$  and  $1.0 \times 10^5 \text{ Pa}$  respectively at a room temperature of  $25^\circ\text{C}$ . Estimate the number of gas molecules trapped in the syringe. (3 marks)

(b) The piston is then pushed in or pulled out to vary  $V$  and  $p$  such that several pairs of readings are recorded. Figure 2.2 shows the graph of  $V$  against  $1/p$  plotted.

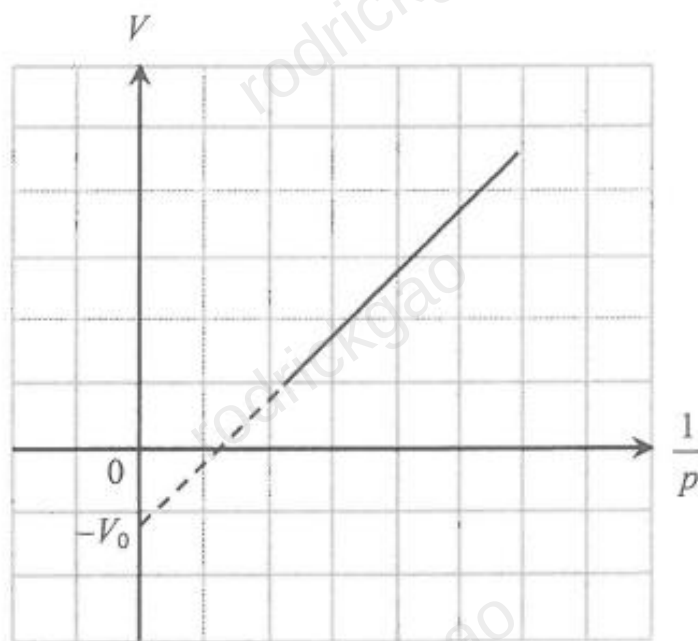


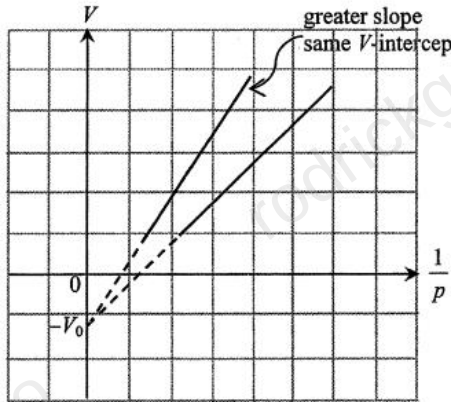
Figure 2.2

(b)(i) State ONE experimental precaution for keeping the gas temperature constant. (1 mark)

(b)(ii) The straight line graph does not pass through the origin but cuts the vertical axis at  $-V_0$  instead. Suggest what  $V_0$  stands for. (1 mark)

(b)(iii) If the experiment is repeated at a higher room temperature using this set-up with the same mass of the same gas, sketch the expected graph in Figure 2.2. (2 marks)

## Official marking scheme

| Solution                                                                                                                                                                                                                                                                                                  | Marks                  | Remarks             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------------------|
| 2. (a) $pV = nRT$<br>$(1.0 \times 10^5)(6.0 \times 10^{-5}) = n(8.31)(273 + 25)$<br>$n = 2.422891 \times 10^{-3} \text{ moles} \approx 2.42 \times 10^{-3} \text{ moles}$<br><br>No. of molecules $= nN_A$<br>$= n \times 6.02 \times 10^{23}$<br>$= 1.458581 \times 10^{21} \approx 1.46 \times 10^{21}$ | 1M<br>1M<br><br><br>1A |                     |
| <i>Alternative method:</i><br>$pV = nRT = \left(\frac{N}{N_A}\right)RT \Rightarrow N = \left(\frac{pV}{RT}\right)N_A$<br>$N = \frac{(1.0 \times 10^5)(6.0 \times 10^{-5})}{(8.31)(273 + 25)} \times (6.02 \times 10^{23})$<br>$= 1.458581 \times 10^{21} \approx 1.46 \times 10^{21}$                     | 1M<br><br>1M<br><br>1A |                     |
|                                                                                                                                                                                                                                                                                                           | 3                      |                     |
| (b) (i) <ul style="list-style-type: none"> <li>- The piston should be pushed or pulled slowly / lightly.</li> <li>- Avoid taking readings immediately after moving the piston.</li> <li>- The syringe should not be grasped by hand too long (when the piston is pushed in or pulled out).</li> </ul>     | 1A<br>Any ONE 1A<br>1A |                     |
|                                                                                                                                                                                                                                                                                                           | 1                      |                     |
| (ii) $V_0$ - volume of gas trapped in the rubber tubing / space connecting the pressure sensor and syringe.                                                                                                                                                                                               | 1A                     |                     |
|                                                                                                                                                                                                                                                                                                           | 1                      |                     |
| (iii)                                                                                                                                                                                                                  | 1A<br>1A               | NO mark for a curve |
|                                                                                                                                                                                                                                                                                                           | 2                      |                     |

(a) Use the ideal gas equation at  $T = 25 + 273 = 298 \text{ K}$ .

$$n = pV/(RT) = (1.0 \times 10^5)(6.0 \times 10^{-5})/(8.31 \times 298) = 2.42 \times 10^{-3} \text{ mol.}$$

$$N = nN_A = (2.42 \times 10^{-3})(6.02 \times 10^{23}) = 1.46 \times 10^{21} \text{ molecules.}$$

(b)(i) Move the piston slowly and wait for thermal equilibrium before taking each reading. This allows any compression heating or expansion cooling to dissipate.

(b)(ii)  $V_0$  is the gas volume in the rubber tubing and connection to the pressure sensor, which is not included in the syringe reading.

(b)(iii) The total volume is  $V + V_0$ , so  $p(V + V_0) = nRT$  and  $V = nRT(1/p) - V_0$ . Draw a straight line with a greater positive slope and the same vertical intercept  $-V_0$ . The increased absolute temperature raises the slope  $nRT$ ; the connecting volume is unchanged.

### Question 3

A person of mass  $m$  stands on a balance inside a lift. The lift goes down from the top of a building at time  $t = 0$  and it reaches the ground at  $t = T$ . The velocity-time ( $v$ - $t$ ) graph of the lift is shown in Figure 3.1. ( $g = 9.81 \text{ m s}^{-2}$ )

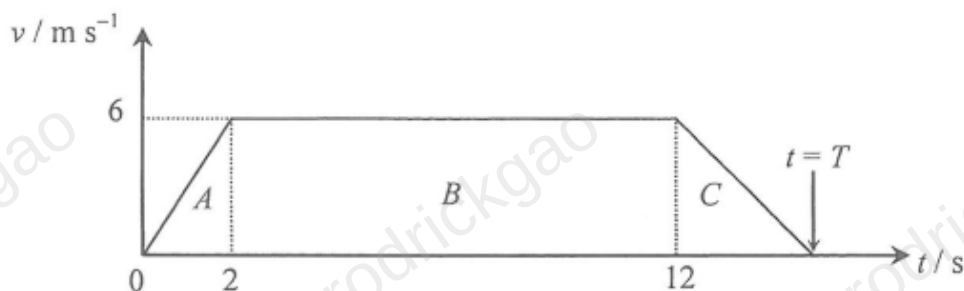


Figure 3.1

(a) Calculate the acceleration of the lift from  $t = 0$  to  $t = 2 \text{ s}$ . (2 marks)

The reading of the balance changes during the person's ride on the lift and the readings registered are 685 N, 569 N and 395 N.

(b) Match these readings with the three stages, A, B and C, of the ride (shown in Figure 3.1). Hence deduce the mass of the person. (3 marks)

(c)(i) Show that  $T = 15 \text{ s}$ . (2 marks)

(c)(ii) Hence estimate the height of the building. (2 marks)

## Official marking scheme

| Solution                                                                                                                                               | Marks          | Remarks                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------|
| 3. (a) $a = \frac{6-0}{2-0}$<br>$= 3 \text{ m s}^{-2}$ (downwards)                                                                                     | 1M<br>1A       | For $g = 10 \text{ m s}^{-2}$ , $m = 56.9 \text{ kg}$                                   |
| (b) A: 395 N      B: 569 N      C: 685 N                                                                                                               | 2              |                                                                                         |
| In stage B, balance reading = weight (Newton's 1 <sup>st</sup> law),<br>$mg = m \times 9.81 \text{ m s}^{-2} = 569 \text{ N}$<br>$m = 58.0 \text{ kg}$ | 1A<br>1M<br>1A |                                                                                         |
| Or In stage A, according to Newton's 2 <sup>nd</sup> law<br>$(569 - 395) \text{ N} = ma = m(3 \text{ m s}^{-2})$<br>$m = 58.0 \text{ kg}$              | 1M<br>1A       |                                                                                         |
| (c) (i) For stage C, by Newton's 2 <sup>nd</sup> law,<br>$F = ma$<br>$(569 - 685) \text{ N} = (58.0 \text{ kg}) a$<br>$a = -2 \text{ m s}^{-2}$        | 3<br>1M        | For $g = 10 \text{ m s}^{-2}$ , $m = 56.9 \text{ kg}$ ,<br>$a = -2.04 \text{ m s}^{-2}$ |
| Thus, $a = \frac{0-6}{T-12} = -2 \text{ m s}^{-2}$<br>$T = 15 \text{ s}$                                                                               | 1M             |                                                                                         |
| (ii) Height $\approx$ displacement of lift = area under graph<br>$= \frac{(12-2)+15}{2} \times 6$<br>$= 75 \text{ m}$                                  | 2<br>1M<br>1A  |                                                                                         |
|                                                                                                                                                        | 2              |                                                                                         |

(a) Downward is positive on the graph.  $a = \Delta v / \Delta t = (6 - 0) / (2 - 0) = 3 \text{ m s}^{-2}$  downwards.

(b) A: 395 N; B: 569 N; C: 685 N. In A the lift accelerates downwards, so the normal force is below  $mg$ . In B acceleration is zero, so the reading equals  $mg$ . In C acceleration is upwards, so the reading exceeds  $mg$ .

Thus  $m = 569 / 9.81 = 58.0 \text{ kg}$ .

(c)(i) In C,  $mg - N = ma$  with downward positive.  $a = (569 - 685) / 58.0 = -2.00 \text{ m s}^{-2}$ .

The graph gives  $(0 - 6) / (T - 12) = -2$ , hence  $T - 12 = 3 \text{ s}$  and  $T = 15 \text{ s}$ .

(c)(ii) Height equals the area under the velocity-time graph:

$$h = \frac{1}{2}(2)(6) + (12 - 2)(6) + \frac{1}{2}(15 - 12)(6) = 6 + 60 + 9 = 75 \text{ m}.$$

## Question 4

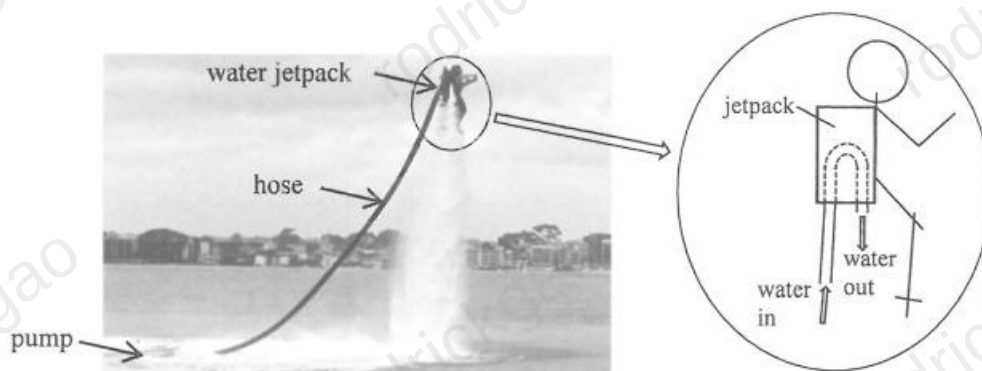
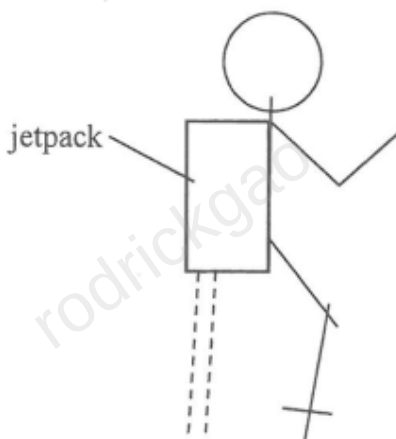


Figure 4.1

A person wears a water jetpack which enables him to stay ‘afloat’ in equilibrium in the air as shown in Figure 4.1. A pump on the sea surface continuously pumps water to the jetpack via a hose and the water is then ejected downwards.

(a) Referring to Figure 4.1, water enters the U-shape hose inside the jetpack with a certain speed and is then ejected out vertically downwards. Use Newton’s law(s) of motion to explain why a lifting force acting on the person is produced. (3 marks)

(b) Draw and label all the forces acting on the person wearing the jetpack as a whole in the free-body diagram below. Neglect the pulling force due to the hose connected to the jetpack. (1 mark)



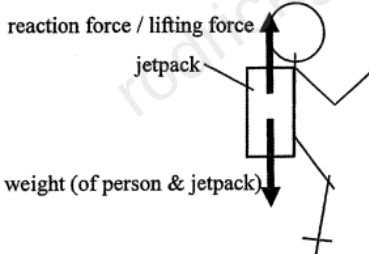
(c) Suppose that water enters the jetpack with a speed of  $10 \text{ m s}^{-1}$  vertically upwards and is then ejected out at the same speed vertically downwards. ( $g = 9.81 \text{ m s}^{-2}$ )

(c)(i) Just by considering the change of momentum of the water, estimate how much water, in kg, has to be ejected per second to provide a lifting force of 1000 N needed. (2 marks)

(c)(ii) Water is pumped to the water jetpack at a height of 7.5 m above sea surface and then ejected from it. By considering the gain in mechanical energy of the water, estimate the minimum output power of the pump. (3 marks)

(d) The person changes to staying ‘afloat’ in equilibrium at a higher position. If the speed by which water enters and is ejected from the jetpack remains the same, would the amount of water ejected per second be greater than, equal to or smaller than the result found in (c)(i)? Explain. (Neglect the weight of the hose.) (2 marks)

## Official marking scheme

| Solution                                                                                                                                                                                                                                                                                                                                                                                                   | Marks                       | Remarks                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4. (a) According to Newton's second law of motion, a (net) force acts on the water so as to change its momentum. (Or magnitude of the force equals the rate of change of momentum of water).<br><br>According to Newton's third law of motion, a force acting downwards on the ejecting water (by the jetpack), the water exerts a reaction (equal but upward / opposite) on the jetpack / person as well. | 1A<br><br>1A<br>1A<br><br>3 |                                                                                                                                                       |
| (b)                                                                                                                                                                                                                                                                                                                       | 1A<br><br>1                 |                                                                                                                                                       |
| (c) (i) $F = \frac{\Delta p}{\Delta t} = \frac{\Delta m}{\Delta t} \times (\vec{v} - \vec{u})$<br>$\frac{\Delta m}{\Delta t} \times (10 - (-10)) \text{ m s}^{-1} = 1000 \text{ N}$<br>$\frac{\Delta m}{\Delta t} = 50 \text{ (kg s}^{-1}\text{)}$                                                                                                                                                         | 1M<br><br>1A<br><br>2       | Accept using kg as the unit.                                                                                                                          |
| (ii) $\left(\frac{\Delta m}{\Delta t}\right)gh + \frac{1}{2}\left(\frac{\Delta m}{\Delta t}\right)v^2$<br>$= (50 \text{ kg s}^{-1})(9.81 \text{ m s}^{-2})(7.5 \text{ m}) + \frac{1}{2}(50 \text{ kg s}^{-1}) \times (10 \text{ m s}^{-1})^2$<br>$= 6178.75 \text{ W or } 6.17875 \text{ kW}$                                                                                                              | 1M<br>1M<br>1A<br><br>3     | Accept using m for $\left(\frac{\Delta m}{\Delta t}\right)$<br><br>For $g = 10 \text{ m s}^{-2}$ , $3750 \text{ W} + 2500 \text{ W} = 6250 \text{ W}$ |
| (d) The same as the same lifting force / (water) jet speed is needed.                                                                                                                                                                                                                                                                                                                                      | 1A<br>1A<br><br>2           |                                                                                                                                                       |

(a) The water's momentum changes from upwards to downwards. By Newton's second law, the jetpack exerts a downward force on the water. By Newton's third law, the water exerts an equal upward force on the jetpack and person, providing lift.

(b) Draw two equal, opposite vertical arrows: upward lifting/reaction force from the water, and downward weight of the person plus jetpack. Do not include the force on the water, since it acts on a different system.

(c)(i) Taking downward positive, the velocity change is  $10 - (-10) = 20 \text{ m s}^{-1}$ .

$F = (\Delta m/\Delta t)\Delta v$ , so  $\Delta m/\Delta t = 1000/20 = 50 \text{ kg s}^{-1}$ .

(c)(ii) Each kilogram gains  $gh$  in potential energy and  $\frac{1}{2}v^2$  in kinetic energy from the nearly stationary sea water.

$P_{\min} = (\Delta m/\Delta t)(gh + \frac{1}{2}v^2) = 50[9.81 \times 7.5 + \frac{1}{2}(10^2)] = 6178.75 \text{ W} \approx 6.18 \text{ kW}$ .

(d) Equal to  $50 \text{ kg s}^{-1}$ . The weight to be supported and the water's velocity change are unchanged, so  $F = (\Delta m/\Delta t)\Delta v$  requires the same mass flow rate. Greater height raises the required pump power, not the flow rate required for this lift.

## Question 5

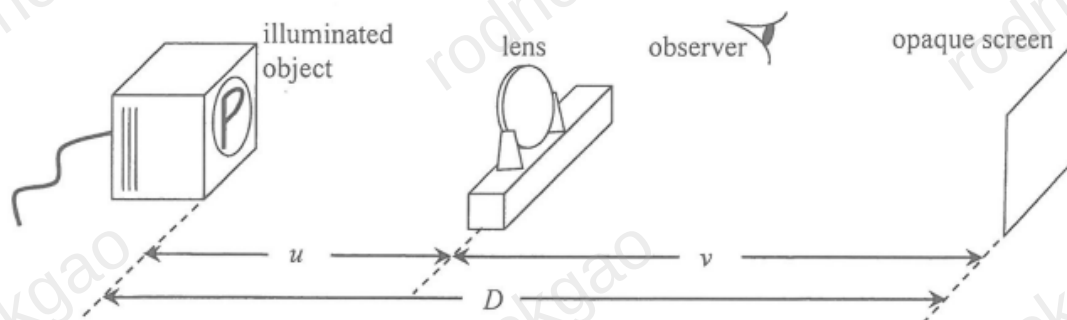


Figure 5.1

Kitty uses the set-up in Figure 5.1 to study the image formation of a lens. The lens is placed at a distance  $u$  from an illuminated object (letter 'P'). An opaque screen is placed at a distance  $D$  from the object so as to capture the image.

- (a)(i) State the kind of lens used. Explain your answer. (2 marks)
- (a)(ii) Sketch the image on the screen seen by the observer. (1 mark)



- (b) The separation  $D$  is varied while the position of the lens is adjusted to form an image once again on the screen. The corresponding object distance  $u$  is obtained for plotting a graph of  $D$  against  $u$  (Figure 5.2).

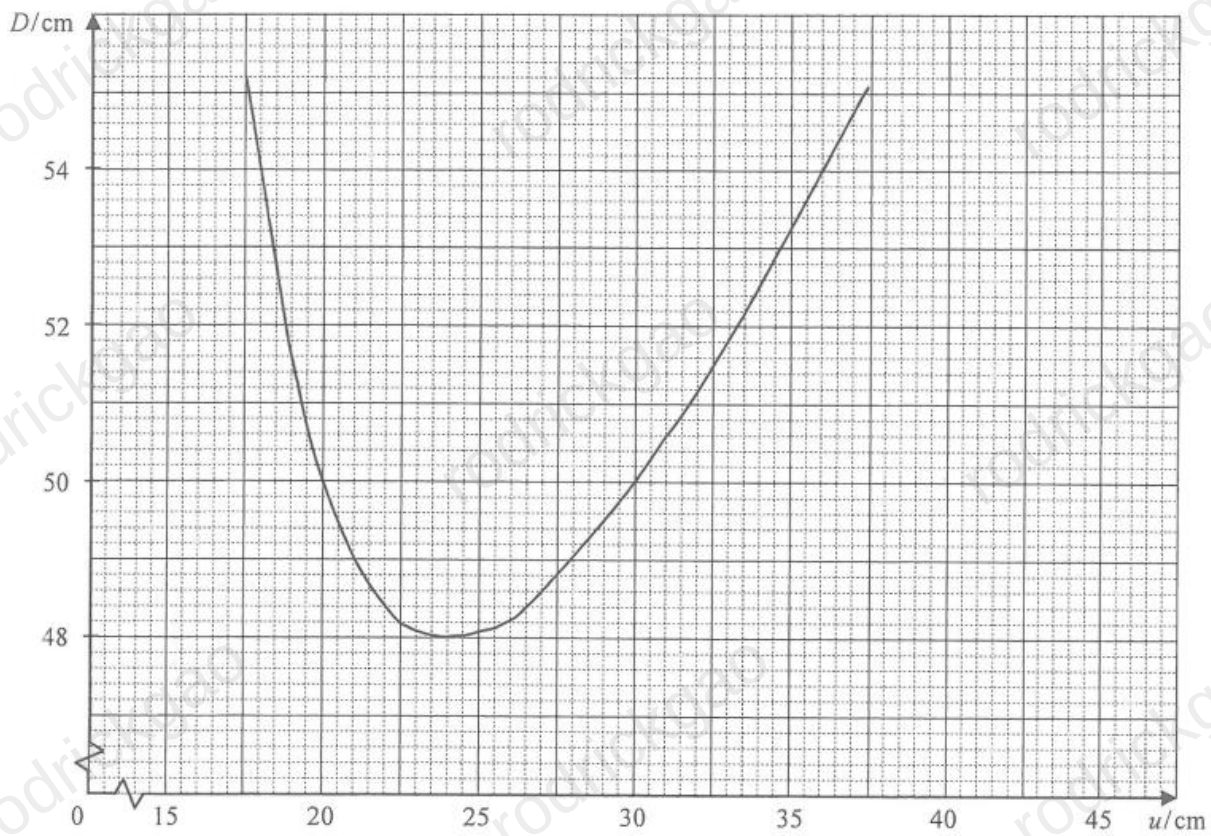
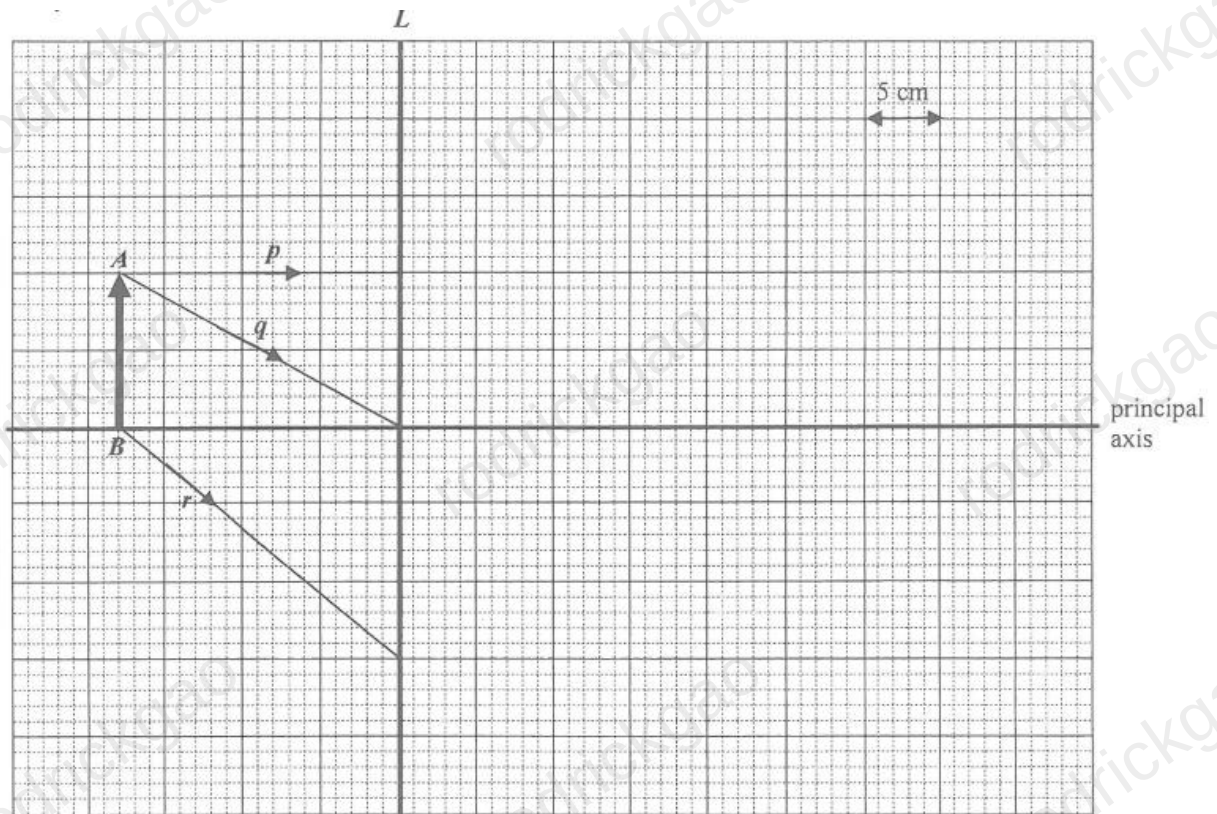


Figure 5.2

**(b)(i)** When the lens is placed at 18 cm from the object, use the graph to find the corresponding separation between the lens and the screen. Hence calculate the magnification of the image. (2 marks)

In the figure below, AB represents the illuminated object which is at 18 cm from the lens L. p, q and r are light rays from AB.

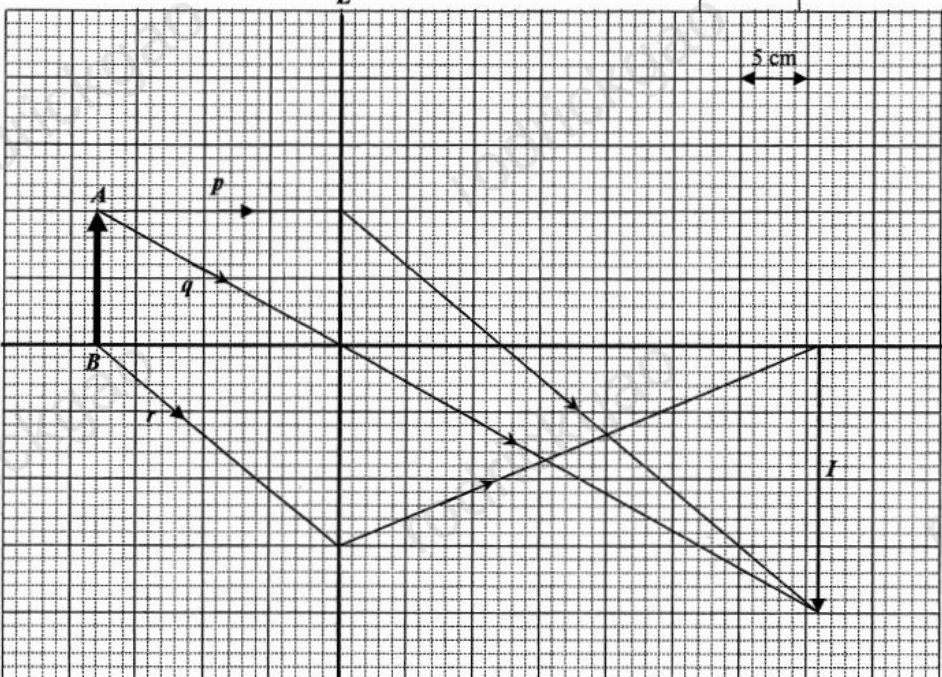


**(b)(ii)** Indicate the image formed by AB (denote it as I) and draw the refracted rays of p, q and r. (3 marks)

**(b)(iii)** Hence find the focal length of the lens. The horizontal scale is 1 cm to 5 cm. (1 mark)

**(b)(iv)** Keeping the object and screen fixed in position, suggest where Kitty should move the lens such that an image can be formed again on the screen. State the ratio (height of this new image)/(height of the original image). (2 marks)

## Official marking scheme

| Solution                                                                                                                                                                                                                                                       | Marks                | Remarks                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|--------------------------------|
| 5. (a) (i) Convex (lens)<br>Only convex lens forms real image (which can be captured by a screen)<br><u>Or</u> Concave lens always forms virtual image (which cannot be captured by a screen).<br><u>Or</u> The image is formed on the other side of the lens. | 1A<br>1A<br>1A<br>1A |                                |
| (ii)                                                                                                                                                                                                                                                           | 2                    |                                |
|  opaque screen                                                                                                                                                                | 1A                   |                                |
| (b) (i)                                                                                                                                                                                                                                                        | 1                    |                                |
| Image distance $v = 54 - 18 = 36 \text{ cm}$ ( $D = 54 \text{ cm}$ )<br>Magnification $= \frac{v}{u} = \frac{36}{18} = 2$                                                                                                                                      | 1A<br>1M/1A          |                                |
|                                                                                                                                                                                                                                                                | 2                    |                                |
|                                                                                                                                                                             |                      |                                |
| (ii)                                                                                                                                                                                                                                                           | 1M<br>1M<br>1M       | 1M for correct position of $I$ |
|                                                                                                                                                                                                                                                                | 3                    |                                |
| (iii)                                                                                                                                                                                                                                                          | 1M/1A                |                                |
|                                                                                                                                                                                                                                                                | 1                    |                                |
| (iv)                                                                                                                                                                                                                                                           | 1M                   |                                |
| <u>Or</u> move the lens 18 cm closer to / towards the screen.<br>Height ratio $= 1 : 4$ .                                                                                                                                                                      | 1M<br>1A             |                                |
|                                                                                                                                                                                                                                                                | 2                    |                                |

**(a)(i)** A convex lens is used: the image can be captured on a screen, so it is real. A single concave lens with a real object produces a virtual image.

**(a)(ii)** Draw the vertically inverted letter as viewed from the observer's side: its vertical stem remains on the left and its loop is at the bottom on the right, as in the official sketch. The viewing side matters.

**(b)(i)** At  $u = 18$  cm, the graph gives  $D = 54$  cm. Thus  $v = D - u = 36$  cm and magnification  $= v/u = 2$ .

**(b)(ii)** Place the inverted image 36 cm to the right of the lens, with twice the object height. Refract p through the far focal point and q straight through the optical centre; they meet at the image of A. Refract r towards the image of B on the principal axis at the same image distance.

**(b)(iii)**  $1/f = 1/18 + 1/36$ , giving  $f = 12$  cm, consistent with the ray diagram (accepted graphical range  $12 \pm 0.5$  cm).

**(b)(iv)** Interchange  $u$  and  $v$ : move the lens 18 cm towards the screen, giving  $u' = 36$  cm and  $v' = 18$  cm. The new magnification is  $\frac{1}{2}$ , compared with 2 originally. New/original image height  $= (\frac{1}{2})/2 = \frac{1}{4}$ , or 1 : 4.

**Question 6 \***

(a) A laser beam is directed perpendicularly towards a double slit of separation  $a = 0.3 \text{ mm}$ . The pattern of bright spots projected on a screen  $1.8 \text{ m}$  away from the slits is shown in Figure 6.1.

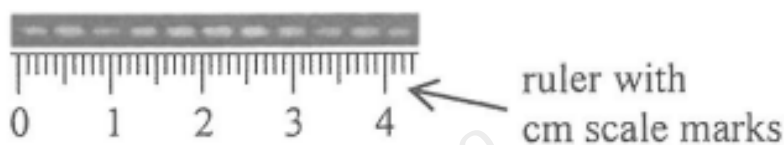
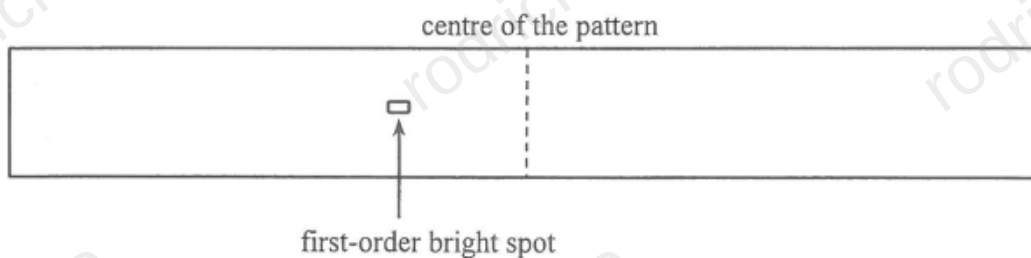
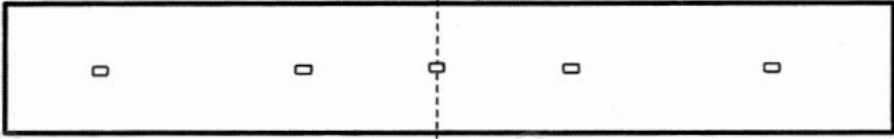


Figure 6.1

- (a)(i) Find the wavelength of the laser beam. (3 marks)
- (a)(ii) Explain why the slit width has to be very narrow in order for the above pattern to be observed. (2 marks)
- (b) The double slit is now replaced by a diffraction grating of 500 lines per mm.
- (b)(i) Find the separation between the central bright spot and first-order bright spot of the pattern on the screen for the same experimental settings. (3 marks)
- (b)(ii) Sketch the pattern, up to the second-order, that you would expect to see on the screen when using this diffraction grating. A first-order bright spot has already been drawn for you. (2 marks)



## Official marking scheme

| Solution                                                                                                                                         |                                                                                                                                      | Marks    | Remarks                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------|
| 6. (a) (i)                                                                                                                                       | $\Delta y = \frac{\lambda D}{a}$                                                                                                     |          |                                                                                                                                          |
|                                                                                                                                                  | $\frac{(4.0-0) \times 10^{-2}}{10} = \frac{\lambda(1.8)}{0.3 \times 10^{-3}}$                                                        | 1M+1M    |                                                                                                                                          |
|                                                                                                                                                  | $\lambda = 6.666667 \times 10^{-7} \text{ m}$                                                                                        | 1A       |                                                                                                                                          |
|                                                                                                                                                  | $\approx 6.67 \times 10^{-7} \text{ m or } 667 \text{ nm}$                                                                           |          |                                                                                                                                          |
|                                                                                                                                                  |                                                                                                                                      | 3        |                                                                                                                                          |
| (ii)                                                                                                                                             | To ensure that the light through the 2 slits diffracts enough to interfere / overlap.                                                | 1A<br>1A | Explanation in terms of formula<br>$\lambda = \Delta y \frac{a}{D}$ is NOT accepted as $a$ is the slit separation given, which is fixed. |
|                                                                                                                                                  |                                                                                                                                      | 2        |                                                                                                                                          |
|                                                                                                                                                  |                                                                                                                                      |          |                                                                                                                                          |
| (b) (i)                                                                                                                                          | $d \sin \theta = m\lambda$                                                                                                           |          |                                                                                                                                          |
|                                                                                                                                                  | $\frac{10^{-3}}{500} \sin \theta = 6.67 \times 10^{-7} \text{ m}$                                                                    | 1M       |                                                                                                                                          |
|                                                                                                                                                  | $\theta = 19.471221^\circ \approx 19.47^\circ$                                                                                       |          |                                                                                                                                          |
|                                                                                                                                                  | Separation between central and first-order bright spot<br>$= 1.8 \tan 19.47^\circ$<br>$= 0.636396 \text{ m} \approx 0.636 \text{ m}$ | 1M<br>1A |                                                                                                                                          |
|                                                                                                                                                  |                                                                                                                                      | 3        |                                                                                                                                          |
| (ii)                                                                                                                                             |                                                                                                                                      |          |                                                                                                                                          |
| <div style="text-align: center;">centre of the pattern</div>  |                                                                                                                                      |          |                                                                                                                                          |
| symmetry about the central bright spot (2 <sup>nd</sup> order shown)                                                                             |                                                                                                                                      | 1A       |                                                                                                                                          |
| separation between 2 <sup>nd</sup> and 1 <sup>st</sup> -order bright spots is larger                                                             |                                                                                                                                      | 1A       |                                                                                                                                          |
|                                                                                                                                                  |                                                                                                                                      | 2        |                                                                                                                                          |

**(a)(i)** There are 10 fringe intervals across 4.0 cm, so  $\Delta y = 0.040/10 = 0.0040 \text{ m}$ .

Using  $\Delta y = \lambda D/a$ ,  $\lambda = a\Delta y/D = (0.3 \times 10^{-3})(0.0040)/1.8 = 6.67 \times 10^{-7} \text{ m} = 667 \text{ nm}$ . Count intervals, not the number of bright spots.

**(a)(ii)** Narrow slits cause sufficient diffraction for light from the two slits to spread and overlap, so interference can occur. Slit width is different from the fixed slit separation  $a$ .

**(b)(i)** Grating spacing  $d = 10^{-3}/500 = 2.00 \times 10^{-6} \text{ m}$ .

For first order,  $d \sin \theta = \lambda$ , hence  $\theta = \sin^{-1}(1/3) = 19.47^\circ$ .

$y_1 = D \tan \theta = 1.8 \tan 19.47^\circ = 0.636 \text{ m}$ . Use  $\tan \theta$  for the screen position; the small-angle approximation is not accurate here.

**(b)(ii)** Draw five spots for orders  $-2, -1, 0, +1$  and  $+2$ , symmetric about the centre. The second-order spots are farther from the first-order spots than the first-order spots are from the centre. Quantitatively,  $y_2 = 1.8 \tan[\sin^{-1}(2/3)] \approx 1.61 \text{ m}$ , so the outward gap is about  $0.97 \text{ m}$ , greater than  $0.636 \text{ m}$ .

## Question 7

(a) The circuits in Figure 7.1 each contain two resistors connected in series with a 6 V battery of negligible internal resistance. The resistors in circuit I are  $10\text{ k}\Omega$  each while those in circuit II are  $100\text{ }\Omega$  each.

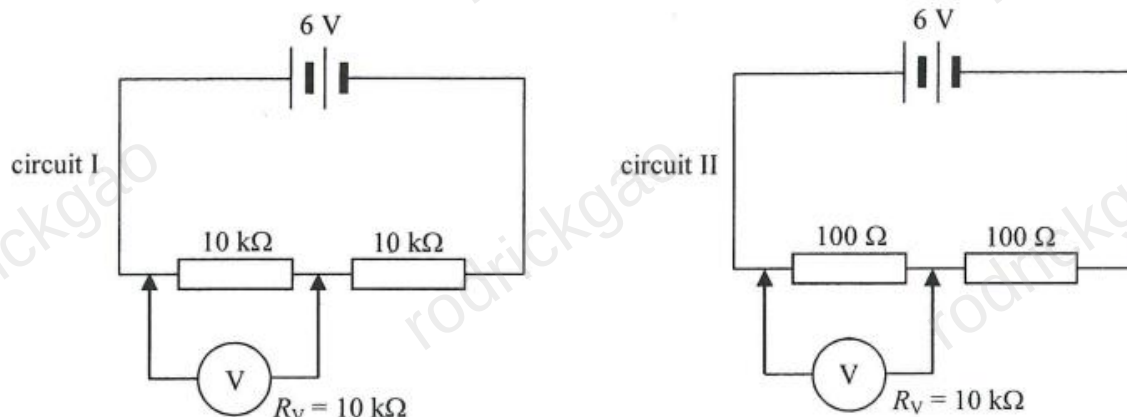


Figure 7.1

A voltmeter of internal resistance  $R_V = 10\text{ k}\Omega$  is used to measure the potential difference across one of the resistors as shown.

(a)(i) What would be the respective voltmeter readings? (3 marks)

(a)(ii) In fact, the potential difference across each resistor before connecting the voltmeter is 3 V in both circuits. Explain why this voltmeter gives a relatively inaccurate value for circuit I. Hence state the general principle of selecting a suitable voltmeter for such measurement. (2 marks)

(b) Circuit III shows a possible method for measuring resistance using a voltmeter and an ammeter. The internal resistances of the voltmeter and the ammeter are  $R_V$  and  $R_A$  respectively and their readings  $V_m$  and  $I_m$  give the measured resistance  $R_m = V_m/I_m$ . The true resistance value of the resistor is  $R$ .

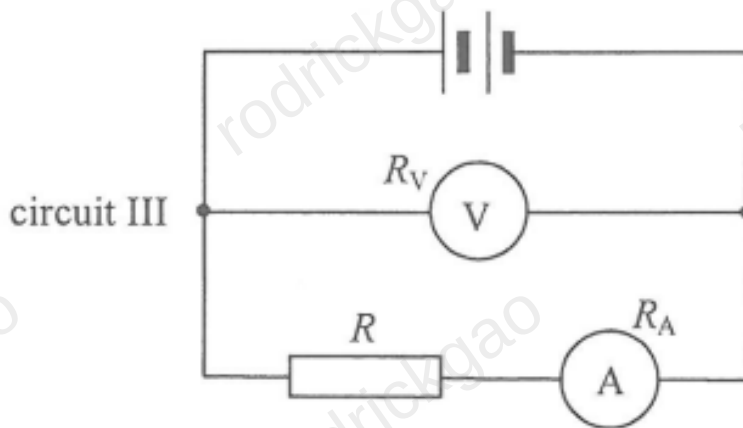


Figure 7.2

(b)(i) State which reading(s),  $V_m$ ,  $I_m$  or both, do(es) NOT give the true voltage across the resistor and/or the true current passing through the resistor. Hence write down an equation relating  $R_A$ ,  $R_m$  and  $R$ . (2 marks)

(b)(ii) Find the percentage error associated with  $R_m$  when measuring the resistance of this resistor.

Given:  $R_V = 10\text{ k}\Omega$ ,  $R_A = 1\text{ }\Omega$  and  $R = 10\text{ }\Omega$ . (2 marks)

## Official marking scheme

| Solution |                                                                                                                                                                                                                   | Marks    | Remarks                                                                                                                              |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------|
| 7. (a)   | (i) $R = 10 \text{ k}\Omega$ (circuit I)                                                                                                                                                                          |          |                                                                                                                                      |
|          | $V = \frac{\left(\frac{1}{10 \text{ k}\Omega} + \frac{1}{10 \text{ k}\Omega}\right)^{-1}}{10 \text{ k}\Omega + \left(\frac{1}{10 \text{ k}\Omega} + \frac{1}{10 \text{ k}\Omega}\right)^{-1}} \times 6 \text{ V}$ | 1M       | 1M for appropriate method in calculating voltage                                                                                     |
|          | $= 2 \text{ V}$                                                                                                                                                                                                   | 1A       |                                                                                                                                      |
|          | $R = 100 \Omega$ (circuit II)                                                                                                                                                                                     |          |                                                                                                                                      |
|          | $V = \frac{\left(\frac{1}{100 \Omega} + \frac{1}{10 \text{ k}\Omega}\right)^{-1}}{100 \Omega + \left(\frac{1}{100 \Omega} + \frac{1}{10 \text{ k}\Omega}\right)^{-1}} \times 6 \text{ V}$                         | 1A       | Note: $100 \Omega$ and $10 \text{ k}\Omega$ in parallel $\approx 99.0099 \Omega$<br>Accept stating that $V$ slightly $< 3 \text{ V}$ |
|          | $= 2.985 \text{ V}$                                                                                                                                                                                               |          |                                                                                                                                      |
|          |                                                                                                                                                                                                                   | 3        |                                                                                                                                      |
|          | (ii) Resistance of circuit / that part of circuit would be lowered / altered significantly when introducing the voltmeter (i.e. loading effect).                                                                  | 1A       |                                                                                                                                      |
|          | Or The resistance of the voltmeter is comparable to the resistance of resistor $R$ .                                                                                                                              | 1A       |                                                                                                                                      |
|          | Resistance of voltmeter should be much higher than the resistance of the part of the circuit under study.                                                                                                         | 1A       |                                                                                                                                      |
|          |                                                                                                                                                                                                                   | 2        |                                                                                                                                      |
| (b)      | (i) $V_m$ does NOT give the true voltage for the resistor.<br>$R_m = R_A + R$                                                                                                                                     | 1A<br>1A |                                                                                                                                      |
|          |                                                                                                                                                                                                                   | 2        |                                                                                                                                      |
|          | (ii) For circuit III<br>$R_m = R + R_A = 10 + 1 = 11 \Omega$<br>percentage error $= \frac{1 \Omega}{10 \Omega} \times 100 \%$<br>$= 10\%$                                                                         | 1M<br>1A |                                                                                                                                      |
|          |                                                                                                                                                                                                                   | 2        |                                                                                                                                      |

(a)(i) Circuit I: the measured  $10 \text{ k}\Omega$  resistor is in parallel with the  $10 \text{ k}\Omega$  voltmeter, giving  $5 \text{ k}\Omega$ . The reading is  $6 \times 5 / (10 + 5) = 2.00 \text{ V}$ .

Circuit II: the parallel combination is  $100 \times 10000 / (100 + 10000) = 99.0099 \Omega$ . The reading is  $6 \times 99.0099 / (100 + 99.0099) = 2.985 \text{ V}$ , slightly below  $3 \text{ V}$ .

(a)(ii) In circuit I the voltmeter resistance is comparable to the resistor, so connecting it substantially reduces the resistance of that branch and changes the voltage division (loading effect). Select a voltmeter whose resistance is much greater than that of the part being measured.

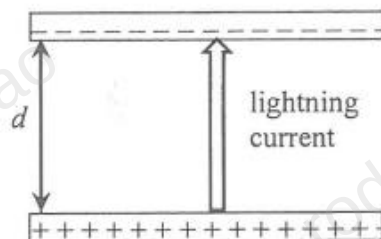
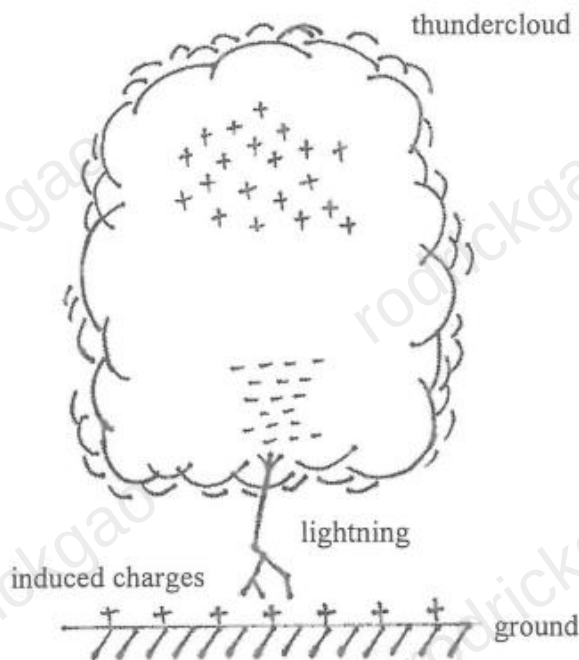
(b)(i)  $V_m$  is not the true voltage across  $R$  alone: it includes the voltage across the ammeter.  $I_m$  is the true current through  $R$ .

$V_m = I_m(R + R_A)$ , hence  $R_m = R + R_A$ .

(b)(ii)  $R_m = 10 + 1 = 11 \Omega$ . Percentage error  $= (R_m - R) / R \times 100\% = 1 / 10 \times 100\% = 10\%$ , an overestimate.

## Question 8

Read the following passage about lightning and answer the questions that follow.



Lightning occurs when charges accumulate in the clouds to such an extent that the electric field in the atmosphere is strong enough to cause the air to lose its insulating properties. The threshold electric field for 'breakdown' to occur is about  $3 \times 10^5 \text{ V m}^{-1}$  above which electrons or ions in the atmosphere can pass through the air between clouds and the ground or between clouds and clouds. The peak current of a typical lightning bolt can reach about 30000 A. How the charges are separated and accumulated in the clouds is not fully understood yet. In most cases, negative charges are at the base of the cloud and positive charges are induced on the ground.

(a)(i) What is the meaning of 'breakdown' in the passage? (1 mark)

\*(a)(ii) The thundercloud's base and the ground can be modeled as two parallel plates with opposite charges. If the negative charges distributed at the cloud's base are about  $d = 2 \text{ km}$  from the ground, find the potential difference between the cloud and the ground when the electric field in the atmosphere just reaches the threshold of 'breakdown'. (2 marks)

A lightning detector having a small square coil inside is placed at point O which is 1.5 km from the lightning bolt. The coil and the lightning's direction are in the same vertical plane as shown. Assume that the lightning current flows vertically upwards to the thundercloud from the ground.

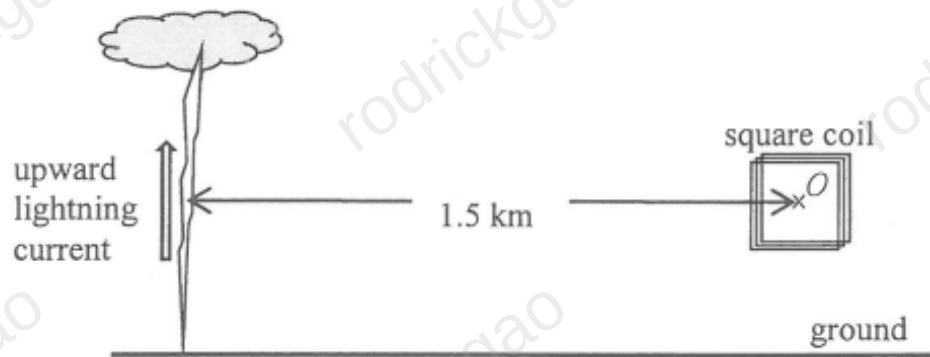


Diagram not drawn to scale

**(b)(i)** State the direction of the magnetic field (to the left / to the right / into paper / out of paper) produced at point O by the lightning current. Estimate the magnetic field strength's peak value at O. (3 marks)

**(b)(ii)** Explain why within the very short duration of lightning an induced current first flows in the coil in a certain direction and then reverses. Your answer should include the directions of the induced current in the coil. (3 marks)

**(b)(iii)** Among the physical quantities related to lightning, electric field in the atmosphere, lightning current and magnetic field due to lightning, suggest which one can be monitored so as to give fore-warning of lightning. Explain your choice. (2 marks)

## Official marking scheme

| Solution |                                                                                                                                                                                                                                                                                                   | Marks | Remarks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------|
| 8.       | (a) (i) - the air loses its insulating properties<br>Or electrons or ions can pass through the air (between clouds and Earth or between clouds and clouds)                                                                                                                                        | 1A    |         |
|          |                                                                                                                                                                                                                                                                                                   | 1A    |         |
|          |                                                                                                                                                                                                                                                                                                   | 1     |         |
|          | (ii) $E = \frac{V}{d}$<br>$V = E d = (3 \times 10^5) \times 2000$<br>$= 6 \times 10^8 \text{ V}$                                                                                                                                                                                                  | 1M    |         |
|          |                                                                                                                                                                                                                                                                                                   | 1A    |         |
|          |                                                                                                                                                                                                                                                                                                   | 2     |         |
|          | (b) (i) magnetic field into paper (due to upward lightning current)<br>$B = \frac{\mu_0 I}{2\pi r}$<br>$= \frac{4\pi \times 10^{-7} \times 30000}{2\pi \times 1500}$<br>$= 4 \times 10^{-6} \text{ T}$                                                                                            | 1A    |         |
|          |                                                                                                                                                                                                                                                                                                   | 1M    |         |
|          |                                                                                                                                                                                                                                                                                                   | 1A    |         |
|          |                                                                                                                                                                                                                                                                                                   | 3     |         |
|          | (ii) When the lightning current is increasing, the induced current flows in the anticlockwise direction so as to oppose the increasing magnetic field (into paper). After reaching maximum, the lightning current is decreasing, the induced current flows in the clockwise / opposite direction. | 1A    |         |
|          |                                                                                                                                                                                                                                                                                                   | 1A    |         |
|          |                                                                                                                                                                                                                                                                                                   | 1A    |         |
|          |                                                                                                                                                                                                                                                                                                   | 3     |         |
|          | (iii) Electric field (in the atmosphere).<br>E-field increases / builds up (to threshold) before lightning happens<br>Or Lightning current and magnetic field only exist during lightning.                                                                                                        | 1A    |         |
|          |                                                                                                                                                                                                                                                                                                   | 1A    |         |
|          |                                                                                                                                                                                                                                                                                                   | 1A    |         |
|          |                                                                                                                                                                                                                                                                                                   | 2     |         |

**(a)(i)** Breakdown means that air loses its insulating property: electrons or ions can pass through it and a discharge can occur.

**(a)(ii)** For the parallel-plate model,  $E = V/d$ . Using the value stated in the passage,  $V = Ed = (3 \times 10^5)(2000) = 6 \times 10^8 \text{ V}$  in magnitude.

**(b)(i)** The right-hand grip rule for an upward current gives a magnetic field into the paper at O.

Approximating the lightning path as a long straight conductor,  $B = \mu_0 I / (2\pi r) = (4\pi \times 10^{-7} \times 30000) / (2\pi \times 1500) = 4.0 \times 10^{-6} \text{ T}$ .

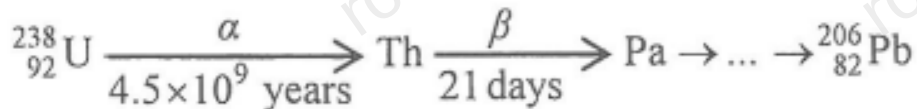
**(b)(ii)** As the lightning current rises, the into-paper flux through the coil increases. By Lenz's law the induced current is anticlockwise, producing an out-of-paper field to oppose that increase.

After the current peaks, the into-paper flux decreases. The induced current becomes clockwise, producing an into-paper field to oppose the decrease. The reversal follows the change in the rate of flux change, not a reversal of the lightning current.

**(b)(iii)** Monitor the electric field in the atmosphere. It builds towards the breakdown threshold before lightning occurs; the lightning current and its magnetic field arise during the discharge and therefore do not provide advance warning.

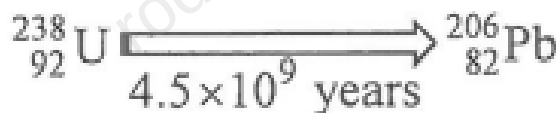
## Question 9

Part of the decay series of uranium-238 (U-238) is shown below. The end product lead-206 (Pb-206) is stable.



(a) When a U-238 nucleus decays to a Pb-206 nucleus, how many  $\alpha$ -particle(s) and  $\beta$ -particle(s) are emitted? (2 marks)

(b) As the first decay in the above chain from U to Th has a half-life much longer than those of subsequent decays, the decay from U-238 to Pb-206 can be simplified to a single decay with half-life  $4.5 \times 10^9$  years:



Suppose that a uranium-bearing rock contains only U-238 and no Pb-206 at the time when it was formed long ago by solidification of molten material. In a particular sample of the rock, it is found that the ratio (number of Pb-206 atoms)/(number of U-238 atoms) is  $2/3$  at present.

(b)(i) Estimate the age of the rock. Assume that all Pb-206 atoms come from the decay of U-238 originally present in the sample and ignore the small number of U-238 atoms which have decayed but have not yet become Pb-206. (2 marks)

(b)(ii) State, with a reason, whether the answer in (b)(i) is an overestimate or an underestimate of the age of the rock if some Pb-206 atoms have actually been lost. (2 marks)

(b)(iii) The graph in Figure 9.1 shows how the number of U-238 atoms in the sample varies with time  $t$  subsequently while  $t = 0$  denotes the present time. On Figure 9.1, sketch a graph to show the variation of the number of Pb-206 atoms in the sample with time. (2 marks)

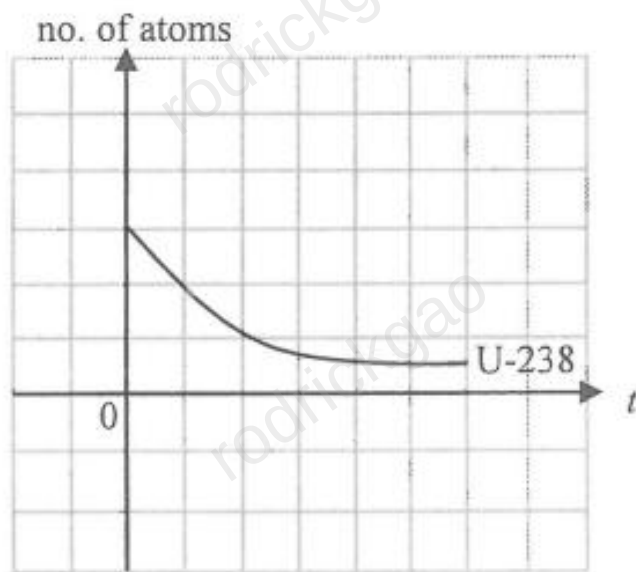


Figure 9.1

## Official marking scheme

| Solution                                                                                                                                                                                                                 | Marks    | Remarks                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------|
| 9. (a) (4) $n_{\alpha} = 238 - 206 \Rightarrow n_{\alpha} = 8$<br>(2) $n_{\alpha} + (-1)n_{\beta} = 92 - 82 \Rightarrow n_{\beta} = 6$                                                                                   | 1A<br>1A |                          |
| (b) (i) $N = N_0 \left(\frac{1}{2}\right)^{t/T_{1/2}}$<br>$\frac{3}{5} = \left(\frac{1}{2}\right)^{t/4.5 \times 10^9 \text{ yr}}$<br>Or $N = N_0 e^{-\lambda t}$ and $\lambda = \frac{\ln 2}{T_{1/2}}$                   | 1M<br>1M |                          |
| $\therefore t = 3.316 \times 10^9 \text{ years} \approx 3.3 \times 10^9 \text{ years}$                                                                                                                                   | 1A       |                          |
| (ii) Answer in (i) is an underestimate.<br>The original number of U-238 atoms should be greater.                                                                                                                         | 1A       |                          |
| $\therefore$ the ratio $\frac{\text{present number of U-238 atoms}}{\text{original number of U-238 atoms}} = \frac{N_t}{N_0}$ is<br>in fact smaller (than $\frac{3}{5}$ ),<br>thus longer time should have been elapsed. | 1A       | Accept 'more U decayed'. |
| (iii)                                                                                                                                                                                                                    |          |                          |
|                                                                                                                                                                                                                          | 2A       |                          |

(a) Each  $\alpha$  decay reduces mass number by 4, so  $n_{\alpha} = (238 - 206)/4 = 8$ . After eight  $\alpha$  decays the atomic number is  $92 - 16 = 76$ . Six  $\beta^-$  decays raise it to 82, so 8  $\alpha$ -particles and 6  $\beta^-$ -particles are emitted.

(b)(i) Present Pb : U = 2 : 3, hence the fraction of original uranium remaining is  $3/(2 + 3) = 3/5$ .

Let  $T = 4.5 \times 10^9$  years be the half-life.  $N/N_0 = (1/2)^{t/T}$ ; therefore  $t = (4.5 \times 10^9) \ln(5/3)/\ln 2 = 3.316 \times 10^9 \text{ years} \approx 3.3 \times 10^9 \text{ years}$ .

(b)(ii) Underestimate. If lead has escaped, more uranium has decayed than the retained lead count indicates. The true original uranium amount is larger, the remaining fraction is smaller, and a longer time has elapsed.

(b)(iii) Here  $t = 0$  is now, not when the rock formed. The lead curve must start at two-thirds of the present uranium count, rise with decreasing slope, and approach the conserved total U + Pb. At each time, the uranium count lost equals the lead count gained. It must not start at the origin.